

Bitcoin Market Sentiment vs Trader Performance Analysis

Dataset	Hyperliquid Historical Trading Data x Bitcoin Fear & Greed Index
Total Trades Analysed	211,224 Closed Trades
Sentiment Regimes	Extreme Fear · Fear · Neutral · Greed · Extreme Greed
Report Type	Exploratory Data Analysis — Hiring Assignment
Date	June 2025

1 Executive Summary

This report investigates the relationship between Bitcoin market sentiment and trader performance on the Hyperliquid perpetual futures platform. By joining 211,224 historical trades with daily Bitcoin Fear & Greed Index classifications, we explore how macro sentiment regimes — spanning **Extreme Fear** to **Extreme Greed** — influence trade volume, profitability, win rates, and directional bias.

Highest Win Rate & Avg PnL	Extreme Greed — 46.49% win rate · \$67.89 avg PnL
Largest Aggregate Profits	Fear regime — \$3.36M total PnL driven by volume & large position sizes
Short-Seller Alpha	Extreme Greed short trades yield \$114.58 avg vs \$10.50 for longs
Long-Buyer Edge	Fear regime long trades yield \$63.93 avg vs \$45.05 for shorts

These findings carry meaningful implications for sentiment-aware trading strategies and risk management frameworks. A sentiment-conditioned approach — aligning position direction and size with regime — appears empirically well-supported by this dataset.

2 Data Description

2.1 Hyperliquid Trading Data

The primary dataset comprises **211,224 closed trades** sourced from Hyperliquid, a decentralised perpetual futures exchange. Each record captures: closed PnL (USD), trade size (USD notional at entry), trade direction (long/short), and a UTC timestamp enabling daily alignment with the sentiment index.

2.2 Bitcoin Fear & Greed Index

The Bitcoin Fear & Greed Index is a composite daily sentiment indicator derived from six weighted signals: price volatility, market momentum, social media sentiment, Bitcoin dominance, trading volume, and Google Trends data. It produces a score from 0-100 mapped to five regimes:

Sentiment Regime	Index Range	Interpretation
Extreme Fear	0 - 24	Panic selling; capitulation risk
Fear	25 - 49	Negative sentiment; risk-off positioning
Neutral	50 - 54	Indecision; low directional conviction
Greed	55 - 74	Positive sentiment; risk-on positioning
Extreme Greed	75 - 100	Euphoria; potential overextension

2.3 Dataset Composition by Sentiment

Sentiment Regime	Trade Count	% of Total
Fear	61,837	29.3%
Greed	50,303	23.8%
Extreme Greed	39,992	18.9%
Neutral	37,686	17.8%
Extreme Fear	21,400	10.1%
Total	211,218	100%

The dataset is skewed toward Fear conditions (29.3%), reflecting Bitcoin’s historical tendency to spend more calendar days in fear regimes relative to euphoric ones. Extreme Fear accounts for only 10.1% of trades, indicating limited market participation during panic periods.

3 Methodology

3.1 Data Integration

Trades were linked to daily sentiment labels by aligning each trade's UTC execution timestamp (truncated to date) with the corresponding Fear & Greed Index classification for that calendar day, producing a single enriched dataset where every trade carries a sentiment label.

3.2 Metrics Computed

Trade Count	Total closed trades per sentiment regime
Average Closed PnL	Arithmetic mean of per-trade realised PnL (USD)
Total Closed PnL	Sum of all realised PnL within the regime (USD)
Win Rate	Proportion of trades with Closed PnL > 0, expressed as %
Avg Trade Size	Mean notional position size per trade (USD)
Directional PnL	Mean PnL segmented by trade direction (Buy vs Sell)

3.3 Analytical Approach

The analysis is primarily descriptive and comparative, using aggregated statistics across sentiment regimes. Findings are interpreted through the lens of behavioural finance — specifically the role of crowd psychology, risk appetite, and positioning in shaping trading outcomes. All monetary values are denominated in USD. No predictive modelling is applied in this iteration; future extensions are noted in the Conclusion.

4 Exploratory Data Analysis

4.1 Average Closed PnL by Sentiment Regime

Sentiment Regime	Avg Closed PnL (USD)	vs Dataset Mean	Rank
Extreme Greed	\$67.89	+\$21.58	1st
Fear	\$54.29	+\$7.98	2nd
Greed	\$42.74	-\$3.57	3rd
Extreme Fear	\$34.54	-\$11.77	4th
Neutral	\$34.31	-\$12.00	5th

Extreme Greed produces the strongest per-trade average (\$67.89), 57% above the Neutral regime. Neutral and Extreme Fear consistently underperform, suggesting poor risk-reward in range-bound and panic-driven markets alike.

4.2 Total Closed PnL by Sentiment Regime

Sentiment Regime	Total Closed PnL	Share of Total
Fear	\$3,360,000	33.8%
Extreme Greed	\$2,720,000	27.4%
Greed	\$2,150,000	21.6%
Neutral	\$1,290,000	13.0%
Extreme Fear	\$740,000	7.4%

Despite Extreme Greed leading per-trade quality, Fear dominates total PnL (33.8% share) through volume and larger average trade sizes. Extreme Fear captures only 7.4% of all profits.

4.3 Win Rate by Sentiment Regime

Sentiment Regime	Win Rate (%)	Trades Won (est.)	Signal
Extreme Greed	46.49%	~18,589	Best
Fear	42.08%	~26,021	Above avg
Neutral	39.70%	~14,962	Below avg
Greed	38.48%	~19,349	Below avg
Extreme Fear	37.06%	~7,930	Weakest

All regimes produce win rates below 50%, confirming profitability arises from payoff asymmetry (winners earning more than losers cost) rather than raw frequency. The 9.4 pp spread between best and worst regimes is meaningful for strategy design.

4.4 Average Trade Size by Sentiment Regime

Sentiment Regime	Avg Trade Size (USD)	vs Extreme Greed
Fear	\$7,816	+\$4,704 (+151%)
Greed	\$5,737	+\$2,625 (+84%)
Extreme Fear	\$5,350	+\$2,238 (+72%)
Neutral	\$4,783	+\$1,671 (+54%)
Extreme Greed	\$3,112	Baseline

Fear produces trade sizes 2.5x larger than Extreme Greed — the most behaviourally revealing metric in the dataset. Experienced traders appear to deploy maximum capital when assets are perceived as undervalued, while euphoric periods attract smaller speculative participants.

4.5 Buy vs Sell Mean PnL by Sentiment Regime

Sentiment Regime	Mean Buy PnL	Mean Sell PnL	Dominant Side	Spread
Fear	\$63.93	\$45.05	BUY	+\$18.88
Greed	\$25.00	\$59.69	SELL	+\$34.69
Extreme Greed	\$10.50	\$114.58	SELL	+\$104.08
Neutral	\$29.23	\$39.46	SELL	+\$10.23
Extreme Fear	\$34.11	\$34.98	~Parity	\$0.87

Highlighted cells mark the dominant directional PnL per regime. The \$104.08 spread in Extreme Greed (sell vs buy) is the largest divergence in the entire dataset — the report's most actionable signal for directional strategy design.

5 Key Findings

Finding 1

Extreme Greed is the highest-quality individual trading environment.

Traders earn the best average PnL (\$67.89), win rate (46.49%), and short-side returns (\$114.58) during Extreme Greed. However, smaller average trade sizes suggest this alpha is being captured by fewer, more selective participants rather than the broad market.

Finding 2

Fear produces the most profitable aggregate regime.

Despite not leading any single per-trade metric, Fear generates \$3.36M in total PnL — \$640K more than Extreme Greed — owing to high volume and large average trade sizes (\$7,816). Experienced traders appear to deploy maximum capital during Fear conditions.

Finding 3	Short sellers are the primary beneficiaries of euphoric markets. The \$114.58 mean sell PnL during Extreme Greed (vs \$10.50 for buys) indicates that while most participants chase longs during bull sentiment, a skilled short-selling cohort extracts outsized returns by fading euphoria. This is the report's most actionable finding.
Finding 4	Long buyers outperform during Fear, consistent with mean-reversion dynamics. Mean buy PnL of \$63.93 during Fear supports the classical contrarian thesis: stepping into negative sentiment periods delivers superior long returns for disciplined traders willing to buy into weakness.
Finding 5	Extreme Fear is the worst risk-adjusted trading environment. Lowest win rate (37.06%), lowest total PnL (\$740K), and near-parity between buy and sell performance all indicate poor edge during panic phases. Reduced trade volume confirms market participants pull back at precisely the wrong moment.
Finding 6	Neutral sentiment consistently underperforms across all metrics. Neutral conditions rank last or second-to-last across average PnL (\$34.31), total PnL (\$1.29M), and win rate (39.70%). Range-bound, directionless markets appear to suppress trading edge for both long and short strategies alike.

6 Strategic Insights

6.1 Sentiment-Conditioned Position Sizing

The data strongly supports scaling position sizes upward during Fear regimes and reducing size during Extreme Greed. The observed trader behaviour (largest sizes during Fear at \$7,816 average) proves rational in hindsight, given Fear's dominance in total PnL. A systematic strategy could formalise this by tying size multipliers to Fear & Greed Index buckets.

6.2 Directional Bias Switching by Regime

The most actionable insight is the clean reversal of directional alpha across regimes: deploy long-biased strategies during Fear (buy PnL leads by \$18.88/trade); deploy short-biased strategies during Extreme Greed (sell PnL leads by \$104.08/trade, the largest spread in the study). A sentiment-conditioned directional filter could materially improve strategy returns by aligning direction with the regime most favourable to that side.

6.3 Reducing Activity in Neutral and Extreme Fear Regimes

Given consistently weak performance across all metrics during Neutral and Extreme Fear conditions, a regime-aware strategy may benefit from reducing activity or moving to flat positions in these windows. The opportunity cost of inactivity during Neutral markets appears low, while Extreme Fear carries both poor win rates and adverse behavioural biases.

6.4 Prioritising Payoff Ratios Over Win Rates

All regimes produce win rates below 50%, confirming profitability is driven by payoff asymmetry rather than hit frequency. Strategy development should prioritise risk-reward management — cutting losses early and allowing winners to extend — over attempting to improve win percentage above 50%.

6.5 Behavioural Finance Interpretation

The patterns align with documented biases: herding and FOMO during Extreme Greed inflate long crowding, depressing buy returns while elevating short-selling value; loss aversion during Extreme Fear causes under-trading precisely when bargain opportunities are greatest; conviction-driven sizing during Fear yields superior aggregate outcomes. A framework that systematically exploits these crowd behaviours is well-supported by this dataset.

7 Conclusion

This analysis of 211,224 trades across five Bitcoin sentiment regimes produces a coherent and actionable narrative: **market sentiment is a meaningful conditioning variable for both trade direction and position sizing decisions.**

The headline insight is the divergence between where individual trading quality peaks (Extreme Greed, driven by skilled short sellers earning \$114.58 average) and where aggregate profits accumulate (Fear, driven by large, directionally aligned long trades averaging \$7,816 in size). These two regimes serve different trader archetypes — the short-seller who fades sentiment extremes and the value-oriented buyer who accumulates during negative sentiment.

Neutral and Extreme Fear regimes offer the least compelling risk-reward profiles and may warrant reduced activity from systematic strategies. A disciplined regime-aware approach — increasing long exposure during Fear, increasing short exposure during Extreme Greed, and reducing size during Extreme Fear and Neutral conditions — aligns well with the empirical patterns observed across all six metrics examined.

Importantly, this analysis is descriptive rather than causal. Future extensions could include multivariate regression controlling for volatility, account-level segmentation to distinguish retail from sophisticated traders, time-series

analysis of regime persistence effects, and out-of-sample backtesting of the proposed strategy rules.

Core Takeaway

The Bitcoin Fear & Greed Index is not merely a descriptive label — it is a **statistically meaningful input** to trading performance that warrants formal incorporation into quantitative trading and risk management frameworks.

Report generated from Hyperliquid historical trade data (211,224 records) joined with the Bitcoin Fear & Greed Index. All PnL figures denominated in USD. Analysis reflects aggregate trader behaviour across the full dataset period and does not constitute financial advice.